Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky solve · reused factor

128 × 128, 1 RHS · f64

Timing direction unclear

CURRENT / BASELINE	1	95% INTERVAL	0.978494623655914 – 1

Cholesky solve · reused factor

256 × 256, 64 RHS · f64

Observed time increase

CURRENT / BASELINE	**1.0877385048482953**	95% INTERVAL	1.0835212691500116 – 1.0905533380292713

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

LU solve · reused factor

256 × 256, 64 RHS · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.980936403843221	0.9784331656759641 – 0.9852986217457886

Thin QR

512 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9980635263938836	0.9794909809735605 – 1.004127710364034

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Thin QR

1024 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	**0.9930393565603598**	95% INTERVAL	0.9869423490620097 – 1.0008656185663485

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	**1.0135542168674698**	95% INTERVAL	0.5107446276861569 – 1.0339157245632067

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.020387359836901	0.990990990990991 – 1.0287769784172662

Thin QR

256 × 64 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0235950780159837	0.9791957376633261 – 1.0424158014687264

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

32 × 32 · 32 × 32 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1	1 – 1

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0010593220338984	0.9968354430379747 – 1.0031712473572938

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	1	95% INTERVAL	0.9957716701902748 – 1.0021231422505308

Matrix multiplication

256 × 256 · 256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	1.0054614964500272	95% INTERVAL	0.9927105831533477 – 1.0194106641721234

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9974789915966387	0.8314212128307188 – 1.0033670033670035

Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.96	0.94 – 0.9795918367346939

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.9978947368421053	0.9895324494068388 – 0.9992997198879552

Independent matrix operands

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9922480620155039	0.9896507115135834 – 1.0025974025974025

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

Timing direction unclear

CURRENT / BASELINE	1	95% INTERVAL	0.9615384615384616 – 1.04

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

Timing direction unclear

CURRENT / BASELINE	1	95% INTERVAL	1 – 1

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9961055819991346	0.9587852494577006 – 1.0378208014407924

Cholesky factorization

256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9976703552708212	0.9940622737146995 – 1.0071553738317758

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.999373094965304	0.9815937955664504 – 1.0053508504807065

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0118906064209274	0.7700320512820513 – 1.256198347107438

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9625962596259626	0.6264762427904422 – 1.0512041150339022

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1	0.9807383627608346 – 1.25705329153605